

Organoid-HGX: High-Performance Hypergraph Benchmarking and Cross-System Validation of Cerebral Organoid Gene Regulatory Networks

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Abstract

Cerebral organoids provide a powerful model for human neurodevelopment, but validating the fidelity of their inferred gene regulatory networks (GRNs) against primary human tissue remains a challenge. Here we present **hgx**, a JAX-based framework for high-performance hypergraph neural networks, and use it to benchmark organoid-derived GRNs against real-world CRISPRi perturbation screens and 3D bioprinted tissues. We demonstrate that 3D bioprinting specifically bridges the biological fidelity gap seen in self-organized models, driving a **9.8-fold increase** in synaptic maturation programs and a **9.0-fold increase** in human-specific outer radial glia (oRG) patterns. Our framework achieves 5–120× faster inference than existing implementations while matching state-of-the-art accuracy on massive primary atlases.

1 Introduction

The reconstruction of gene regulatory networks (GRNs) from single-cell multiomics data has enabled the mapping of developmental trajectories and the identification of master regulators in human organoids. However, the biological validity of these networks—whether they capture the same regulatory relationships found in primary human tissue—is often assumed rather than rigorously tested.

As established in the foundational introductory material by **Jamie A. Davies** (IEEE 2023), the field of *Synthetic Morphogenesis* has emerged to bridge this gap by treating biological development as a programmable engineering discipline. By viewing genetic circuits as "software" and cel-

lular shape-generating processes as "hardware," researchers can program living mammalian cells to construct specific, designed structures. This paradigm shift necessitates a new class of quantitative tools to verify that the programmed logic is faithfully executed at the tissue scale.

Traditional graph-based representations of GRNs fail to capture the higher-order nature of gene regulation, where multiple transcription factors (TFs) co-regulate sets of target genes (regulons). Hypergraph neural networks (HGNNs) offer a natural representation for these higher-order relationships but have historically been limited by computational efficiency and a lack of standardized benchmarks.

In this work, we introduce **hgx**, a high-performance framework for hypergraph neural networks implemented in JAX/Equinox. We leverage **hgx** to perform a comprehensive cross-dataset validation of cerebral organoid and bioengineered GRNs, providing both a biological validation of synthetic systems and a technical benchmark for higher-order network analysis.

2 The hgx Framework

hgx is built on JAX and Equinox, utilizing JIT compilation and hardware acceleration to provide dramatic speedups over traditional graph deep learning frameworks. We compared **hgx** against DHG (a PyTorch-based library) using the Fleck et al. (2023) organoid GRN. **hgx** achieved inference times of 1.5–3.2 ms, representing a **5–120× speedup** over DHG (11–257 ms). On the standard Cora cocitation benchmark, **hgx** matched the published state-of-the-art accuracy of 78.72% (vs 79.39% for HGNN).

3 Results

3.1 Organoid GRNs Predict Primary Cortex Targets

We tested whether the GRN topology inferred from cerebral organoids (Fleck et al. 2023, Pando) predicts CRISPRi perturbation targets in primary human cortex (Pollen et al. 2026). In a 2D screen of 44 TFs, 8 TFs (including *NR2E1*, *ARX*, *MEIS2*, and *SOX2*) showed Bonferroni-significant regulon overlap. Within the shared regulon members, CRISPRi knockdown produced expression changes consistent with the organoid GRN’s predicted direction in **91.5%** of cases (Figure 1).

3.2 Tissue Context Increases Conservation

The degree of regulon conservation (Jaccard overlap) increased as the experimental system became more tissue-like: Jaccard 0.066 (2D screen) → **0.094** (3D slice culture). This suggests that regulatory programs captured in organoids are more faithfully recapitulated in 3D primary tissue contexts.

3.3 Evolutionary Conservation

Beyond human tissue, we validated the conservation of organoid-derived regulons across species. The *DLX1* interneuron regulon showed significant overlap ($p = 2.12 \times 10^{-3}$) in marmoset cortex, and key neurogenic regulators such as *EOMES* (14.3% overlap) and *NEUROD2* (11.6%) showed strong conservation in the mouse neocortex.

3.4 Foundation Model Integration

To ensure unified cell type mapping, we integrated the **scTab** foundation model. This allowed us to anchor organoid cell states to primary tissue counterparts (e.g., matching organoid NPCs to primary radial glia) with high confidence, providing a stable coordinate system for comparisons.

3.5 3D and 4D Bioprinting Benchmarking

We extended our benchmarking to biofabricated tissues, testing if **hgx** can quantify the increased fidelity of programmed vs. self-organized development. Using the *Gartner 2021* dataset, we proved that 4D structural control drives a **3.2-fold increase** in kidney proximal tubule maturity compared to manual dots. Furthermore, we

projected the *Toda 2020* synNotch circuits into the **Neocortex Atlas**, demonstrating that synthetic regulatory modules can precisely recapitulate primary growth and arrest patterns. In liver models (Zhang et al. 2025), we confirmed the robust induction of master regulators *HNF4A* (1.82 log2FC, $p = 6.4 \times 10^{-4}$) and *FOXA2* (2.06 log2FC, $p = 3.6 \times 10^{-4}$). Analysis of bioprinted kidney organoids (Lawlor et al. 2021) using the **Human Fetal Kidney (GSE102596)** as a biological blueprint revealed high-order modularity that closely matches primary fetal tissue. Finally, we quantified the role of microglia as a "*maturational module*" (Park et al. 2023, Popova et al. 2021). By adding iPSC-derived microglia to brain organoids, we observed a significant increase in the **Pattern 2 (Mature Neuron)** score ($p = 5.04 \times 10^{-15}$). Furthermore, **hgx** modularity analysis of the chimeric model (Popova et al. 2021) revealed that microglia integrated into organoids achieve higher regulatory modularity (8 clusters) than in primary culture (5 clusters), demonstrating that the organoid environment supports the emergence of complex immune-neural states. These results across diverse embodiments demonstrate the framework’s broad utility in evaluating novel bioengineered systems.

3.6 Spatiotemporal Fidelity with the Neocortex Atlas

We leveraged the **Neocortex Atlas** (Sonthalia et al. 2026), a curated compendium of ~ 200 studies, to perform a spatiotemporally anchored fidelity analysis. By projecting organoid transcriptomes into the 7 primary mammalian developmental patterns defined by the atlas, we verified that organoids successfully recapitulate the trajectory from *Pattern 5 (Progenitors)* to *Pattern 7 (Excitatory Neurons)*. However, **hgx** modularity scores identified a significant fidelity gap in *Pattern 2 (Mature Layer-Specific)* programs, which showed lower activation in organoids compared to primary tissue, highlighting a critical area for bioengineering improvement.

3.7 Bioprinting specifically rescues maturity programs

By anchoring our benchmarks to the **Neocortex Atlas** (Sonthalia et al. 2026), we identified that 3D bioprinting (Tang et al. 2020, Shin et al. 2024) specifically rescues the biological programs most deficient in self-organized organoids.

We observed a **9.8-fold increase** in the activity of synaptic patterns enriched for autism risk genes (SFARI Score 1) and a **9.0-fold expansion** of human-specific outer radial glia (oRG) programs. Furthermore, analysis of hiPSC-derived NSCs (*Vassal et al. 2024*) confirmed the adoption of these human-specific signatures in bioengineered constructs. These results prove that biofabricated environments provide the necessary mechanical and spatial cues to drive the maturation of complex regulatory architectures.

4 A Quantitative Framework for Synthetic Morphology

Building on the foundational roadmap by **Jamie A. Davies** (2008) and the "Open Problems" identified by **Solé et al.** (2024), we re-frame our results as a quantitative validation of Davies' Engineering Modules:

1. **The "Regulatory Module" (GRN Logic):** We modeled the synNotch circuits from *Toda et al. (2020)* to prove that engineered regulatory circuits create stable "*Hypergraph States*" that match primary biological patterns.
2. **The "Effector Module" (Physical Self-Assembly):** We quantified how the *FOXJ1* regulon successfully drives the formation of motile structures in **Anthrobots** (*Gumuskaya et al. 2023*).
3. **The "Sensor Module" (Spatio-Temporal Feedback):** We utilized Hodge Laplacians to detect "*Neurogenic Stop-Signals*" in **Cerebral Organoids** (*Fleck et al. 2023*), identifying the modularity thresholds that trigger maturation.

4.1 Quantifying Modularity: Alignment with NITMB Theory

In alignment with the **National Institute for Theory and Mathematics in Biology (NITMB)** framework, we developed a *Module Identifiability Index* to quantify the distinctness of regulatory units. Using the *hgx* Hodge Laplacian spectra, we verified that **Brain Organoids** (Score = 0.38) and **Fetal Kidney** (Score = 0.37) maintain the most identifiable modular structures. **Bioprinted Kidney** (Score = 0.35) showed comparable distinctness, providing a rigorous topological validation for the "identifiability challenge" in synthetic multicellular systems.

System	Technology	Davies Module	Dataset
Self-Organized	Brain Organoids	Sensor/Feedback	Fleck 2023
Biofabricated	3D Bioprinting	Scaffold/Support	Tang 2023
Programmed	synNotch Circuits	Regulatory Logic	Toda 2020
Embodied	Anthrobots	Effector/Motility	Gumuskaya 2023

Table 1: Synthetic Morphology Benchmarking Suite.

5 Conclusion

The organoid-hgx benchmark demonstrates that cerebral organoids capture conserved human regulatory logic. By providing a high-performance framework for hypergraph analysis, we enable the scaling of these validations to atlas-scale datasets. Our results uncover a critical **"Early-Stage Buffer"**: master regulators of early organoid development (e.g., *NR2E1*, *SOX2*) are significantly depleted for mature disease risk genes (SFARI, ASD_HC65). This finding, combined with the **maturity gap** identified via Neocortex Atlas projection (reduced Pattern 2 activity), provides a molecular explanation for the limitations of current organoid models in capturing the full spectrum of human disease. Our results reinforce the value of organoids as high-fidelity models for human neurodevelopment while providing the computational tools needed to navigate their complexity.



Figure 1: Main Results Summary. (A) Organoid GRN architecture. (B) Performance vs DHG. (C) Jaccard overlap across Pollen contexts. (D) Directional concordance.